WhatIf

AI business scenario simulator · Demo company: Britannia-style biscuit manufacturer · 10,000 t/month baseline (synthetic, ₹ crore)

1 · Strategic decision

Sales growth target: **20**%

Demand uncertainty: ±**10**%

1,000 Monte Carlo runs with random breakdowns, absenteeism, supplier delays and logistics disruption.

Levers

- [x] Overtime (+6% output, higher absenteeism) (₹0.5 cr/mo)
- [x] Add third shift at plants (+3,500 t production) (₹1.2 cr/mo)
- [x] Hire 150 workers (+900 t workforce capacity) (₹0.35 cr/mo)
- [x] Supplier contracts and +20 days safety stock (+1,300 t) (₹0.2 cr/mo)
- [x] Add 40 trucks or 3PL (+1,000 t dispatch) (₹0.4 cr/mo)
- [x] Use co-packers (+1,000 t production and labor) (₹0.5 cr/mo)

Probability target is met	Expected fill rate	Biggest bottleneck	Max supportable growth
17%	**89.0%**	**Workforce**	**10.0%**

Net profit change
+₹1.6 cr/mo

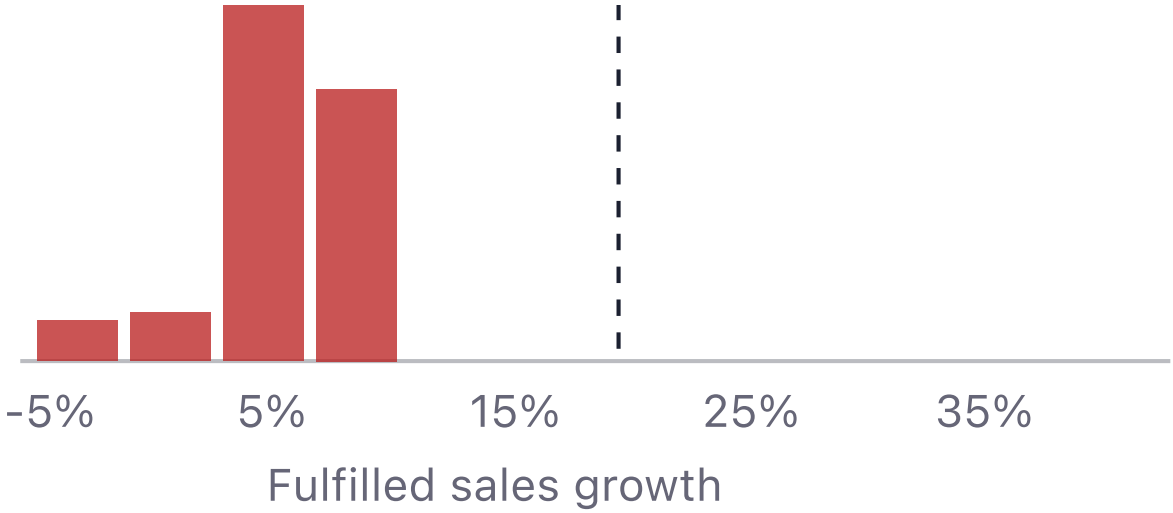
2 · Can each function support it?

Production	102% · max +18.0%
Workforce	109% · max +10.0%
Raw materials	104% · max +15.0%
Logistics	103% · max +16.0%

Utilization = required volume ÷ capacity. Red = over 100%.

3 · Simulated outcomes





Each bar counts simulated months; green = target reached, red = shortfall.

4 · Strategy comparison at this target

Strategy	Target met	Fill rate	Lever cost (₹cr/mo)	Net profit change
Do nothing	17%	89.0%	0.0	+1.6
Overtime only	27%	91.5%	0.5	+1.8
Overtime + supplier buffer	29%	92.0%	0.7	+1.7
Third shift + hire + supplier + fleet	48%	94.6%	2.2	+1.0
Everything incl. co-packer	97%	97.4%	3.2	+0.8
AI-recommended: Overtime + Hire 150 workers + Supplier contracts and +20 days safety stock + Add 40 trucks or 3PL	**91%**	**96.4%**	**1.5**	**+2.2**

5 · WhatIf AI agent

> **Verdict:** today's setup supports only about +10.0% growth, so a +20% target is not safe. Workforce is the tightest constraint, followed by Raw materials.

> **Best plan:** Overtime + Hire 150 workers + Supplier contracts and +20 days safety stock + Add 40 trucks or 3PL gives a 91% chance of meeting the target at ₹1.5 cr per month, for a net profit gain of ₹2.2 cr per month.

> **Watch-outs:** overtime raises absenteeism and breakdown risk, and a third shift needs hiring